Heights of positive weighted elliptic traces

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Abstract

A positive weighted trace of several elliptic abscissas controls all their indices, even when the weights vary through rational squares. The bound is linear in the heights of the weights and the resulting trace. The additive constant is ineffective. The result includes half-integral indices, repeated points, negative indices, and finite torsion points. The proof checks the exceptional translates and projective embedding in the published family height theorem. A separate quadratic-reciprocity lemma shows that the existing Gaussian prime conditions make the chosen half-generator divisible by two in both residue groups. Together these results permit two point differences to replace two stored doubled points in the quartic construction.

1 The curve, indices, and statement

Put $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, $d = 3 - \rho$, $d' = 3 + \rho$, $D = d^2$, and

$$E : y^2 = x^3 - Dx, \qquad R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0.$$

Retain the arithmetic hypothesis $2E(K) = \mathbb{Z}P$, with P of infinite order. Every $Q \in E(K)$ has a unique expression $Q = [m]R_0 + T_0$, where $m \in \mathbb{Z}$ and $T_0 \in E[2](K)$. Indeed, $2Q = mP$ implies $2(Q - mR_0) = O$, and uniqueness follows from non-torsion. Define its index by $\nu(Q) = m/2$. Write h for absolute logarithmic Weil height, and $\widehat{h}$ for elliptic canonical height. Thus $\widehat{h}(Q) = \nu(Q)^2\widehat{h}(P)$.

Theorem 1.1. *Fix a positive integer s and fixed polynomials $w_1, \dots, w_s \in \mathbb{Q}[Z]$. There are constants $A > 0, J \geq 0$ such that, for every $r \in \mathbb{Q}$ with $w_i(r) > 0$ for all i, and every $Q_i \in E(K) \setminus \{O\}$, the number*

$$S = \frac{1}{4} \operatorname{Tr}_{K/\mathbb{Q}} \sum_{i=1}^{s} w_i(r) x(Q_i)$$

satisfies

$$\max_i |\nu(Q_i)|^2 \leq A\big(h(r) + h(S)\big) + J.$$

The constants depend only on the fixed curve, subgroup and polynomials, not on r or the points. The argument does not make J effective. In particular the conclusion applies to any prescribed assignment of weights 1 and r^2, on $r \neq 0$.

This is a theorem relative to the displayed group hypothesis, not a new *abc* assumption. The analytic input to its proof is the published Theorem 1.1 of von Buhren [1]. No prime-pattern or pole-realization hypothesis enters this height argument. The positivity is imposed at the actual rational parameter, not at every algebraic point of its projective parameter space.

2 Independence of the conjugate generators

Let $E_0 : Y^2 = X^3 - X$. Over a fixed field containing $K(i, \sqrt{d}, \sqrt{d'})$, use the pointed isomorphisms

$$(x, y) \longmapsto (x/d, y/d^{3/2}), \qquad (x, y) \longmapsto (x/d', y/(d')^{3/2}).$$

Let R, R' be the respective images of R_0 and its conjugate.

Lemma 2.1. *The points R, R' are linearly independent modulo torsion over* $\mathrm{End}(E_0) = \mathbb{Z}[i]$.

Proof. The square classes of d, d' are independent in $K(i)^\times / K(i)^{\times 2}$. For completeness, an element of K is a square in $K(i)$ only if it or its negative is a square in K: the imaginary part of $(a+bi)^2$ forces $ab = 0$. Neither d nor $-d$ is a square in K, because its norm is 7, not a rational square. The same holds for d'. Their product is 7. If $(a + b\rho)^2 = 7$ with $a, b \in \mathbb{Q}$, then $ab = 0$, giving either $a^2 = 7$ or $2b^2 = 7$, both impossible. The negative of 7 is not a square in the real field K. This proves the claimed independence.

There is consequently an automorphism of the fixed compositum fixing $K(i, \sqrt{d})$ and sending $\sqrt{d'}$ to $-\sqrt{d'}$. It fixes R and negates R', while fixing every endomorphism in $\mathbb{Z}[i]$. If $\alpha R + \beta R'$ is torsion, subtraction of its image gives $2\beta R'$ torsion. A nonzero elliptic endomorphism is an isogeny; its inverse image of torsion consists of torsion points. Since R' is non-torsion, $\beta = 0$, and then $\alpha = 0$. □

3 Noncancellation at poles

Consider an actual tuple of points of E_0^{2k} whose first block has coordinates $[m_j]R + T_j$ and whose second has coordinates $[m_j]R' + T'_j$, where $m_j \neq 0$ and the T_j, T'_j are torsion. Fix positive rational weights a_j. The function to be considered is

$$F = \frac{d}{4} \sum_{j=1}^{k} a_j x_j + \frac{d'}{4} \sum_{j=1}^{k} a_j x'_j + c,$$

where c is any fixed constant.

Lemma 3.1. *No positive-dimensional abelian translate through this tuple, meeting the domain of F, has constant restriction of F.*

Proof. Any positive-dimensional abelian subvariety of a power of E_0 contains an elliptic subvariety isogenous to E_0. This follows from complete reducibility and the fact that every simple constituent of a subvariety of E_0^{2k} is isogenous to E_0. Compose an isogeny from E_0 to such a subvariety with the inclusion. The resulting nonzero direction has coordinate endomorphisms $\alpha_j, \alpha'_j \in \mathbb{Z}[i]$. If a translate of the larger subvariety had constant F, the translate in this direction through our tuple would also have constant F.

At a pole of a nonconstant coordinate in the first block, two such coordinates i, j can have a common pole only if $[\alpha_j]Q_i - [\alpha_i]Q_j = O$. Non-torsion then gives $m_i\alpha_j = m_j\alpha_i$. In a local analytic parameter on E_0, the principal part of x at the identity is z^{-2}. Thus the coefficient of the common double pole, summed over all first-block coordinates having that pole, is

$$\frac{d}{4\alpha_i^2} \sum_j a_j (m_i/m_j)^2 \neq 0.$$

The sum is a strictly positive rational number. This argument allows negative or repeated indices and arbitrary torsion translates.

A first-block and a second-block nonconstant coordinate cannot have a common pole: the same elimination would give a relation $m_i\alpha'_j R - m_j\alpha_i R'$ modulo torsion, contradicting Lemma 2.1. The identical positive-sum argument holds within the second block. At least one direction coordinate is nonzero, so at least one nonconstant summand has a pole, and it cannot cancel. Consequently the restriction is not constant. □

4 Application of the family height theorem

Proof of Theorem 1.1. First separate the finitely many possibilities for those Q_i whose index is zero. Such points are finite two-torsion points; freeze their coordinates and omit them from the abelian ambient space. Their contributions to the function are fixed polynomials in r. There are only finitely many choices of omitted index sets and torsion points. If all points are omitted, the left side of the asserted bound is zero. Otherwise let $k \geq 1$ denote the number of remaining indices.

On E_0^{2k}, with an additional parameter r, form the rational function

$$F_r = \frac{d}{4} \sum_j w_j(r) x_j + \frac{d'}{4} \sum_j w_j(r) x'_j + c(r),$$

where $c(r)$ accounts for the frozen torsion coordinates. Let $\mathcal{V}$ be the integral reduced closure of its graph in $E_0^{2k} \times (\mathbf{P}^1_r \times \mathbf{P}^1_S)$. This closure is integral because the graph over its nonempty rational-function domain is integral. All actual tuples under consideration lie in that domain: r and all abscissas are finite.

Fix the finite-rank group generated by R in the first-block factors, R' in the second-block factors, and the finitely many needed torsion coordinates. It is independent of r and of the integer indices. At an actual parameter with all weights positive, Lemma 3.1 excludes every positive-dimensional abelian translate in the fiber through our tuple. A translate in the projective graph fiber through a domain point would meet the domain densely and make the displayed rational function constant there; thus graph boundary components do not evade the lemma.

We check all additional hypotheses of [1, Theorem 1.1, p. 308]. Embed each E_0 by $\mathcal{O}(3O)$ and its product by Segre. For $n \geq 1$, the plane-cubic sequence

$$0 \to \mathcal{O}_{\mathbf{P}^2}(n-3) \to \mathcal{O}_{\mathbf{P}^2}(n) \to \mathcal{O}_{E_0}(3nO) \to 0$$

has surjective restriction of global sections, since $H^1(\mathbf{P}^2, \mathcal{O}(n-3)) = 0$. Tensoring these surjections and using Segre gives the source's condition (1.1) for every $n \geq 1$. The resulting line bundle on the abelian variety is ample and symmetric. Embed the parameter surface by $\mathcal{O}(1,1)$, whose height is $h(r) + h(S)$ up to a fixed bounded term. The ambient parameter variety is projective and integral. Every point of the chosen finite-rank group belongs to the source set $C(\Gamma, \epsilon)$ by taking the error point to be zero. The source theorem therefore bounds its canonical product height by a fixed multiple of $h(r) + h(S)$ plus a fixed additive constant.

That canonical height is a fixed positive multiple of $2\sum_j \widehat{h}(Q_j)$, by pointed-isomorphism and Galois invariance. As $\widehat{h}(P) > 0$, it bounds the largest squared index. Take the largest constants over the finitely many frozen-torsion cases. All constants are now independent of r and the points. The cited theorem does not make the additive constant effective. □

Corollary 4.1. *For finite $Q_1, Q_3 \in E(K)$ and nonzero doubled points $Q_2 = [a]P, Q_4 = [b]P$, with $a, b \in \mathbb{Z} \setminus \{0\}$, put $\nu(Q_1) = n$, $2\nu(Q_3) = t \in \mathbb{Z}$ and*

$$T = \frac{1}{4} \operatorname{Tr}(x(Q_1) + x(Q_3) + r^2(x(Q_2) + x(Q_4))).$$

There are fixed A, J with

$$\max(|n|, |a|, |t|, |b|)^2 \leq A(h(r) + h(T)) + J \qquad (r \neq 0).$$

If $r > 1$, then $T > 0$.

Proof. The factor two in t only enlarges the height constant by four. At either real embedding every finite elliptic point has $x \geq -d$, since $x(x-d)(x+d) \geq 0$. A nonzero doubled point has square abscissa and is non-torsion, hence $x > d$. Summing conjugate embeddings gives $\operatorname{Tr}(x(Q_1) + x(Q_3)) \geq -12$ and $\operatorname{Tr}(x(Q_2) + x(Q_4)) > 12$, proving $4T > 12r^2 - 12$. □

5 Divisibility of the half-generator at selected primes

Lemma 5.1. *Let $p = A^2 + B^2$ be prime, with $A \equiv 9 \pmod{16}$ and $B \equiv 0 \pmod{16}$. Then R_0 and its conjugate belong to $2E(\mathbf{F}_p)$ in their respective reductions. Consequently $\#E(\mathbf{F}_p)/4$ annihilates the reduction of R_0 whenever E has good reduction, and the analogous statement holds for the conjugate curve.*

Proof. We have $AB \neq 0$, $\gcd(A, B) = 1$ and $p \equiv 1 \pmod{8}$. In $\mathbf{F}_p$ put $i = A/B$. Then $i^2 = -1$. Quadratic reciprocity gives $(B/p) = 1$: the 2-part and sign are squares, and for every odd prime $q \mid B$ one has $(p/q) = (A^2/q) = 1$ and $p \equiv 1 \pmod{4}$. Likewise, using Jacobi symbols if $A + B$ is composite,

$$\left(\frac{A+B}{p}\right) = \left(\frac{p}{|A+B|}\right) = \left(\frac{2}{|A+B|}\right) = 1.$$

Here $p \equiv 2A^2 \pmod{|A+B|}$, and $A + B \equiv 1 \pmod{8}$; absolute value does not change the symbol of 2. Thus $2+2i$ is a square. Its product with $2-2i$ is 8, a square modulo p, so the latter is a square as well. Choose their square roots u, v. Then $uv = \pm 2\sqrt{2}$, and $(u \pm v)^2/4 = 1 \pm \sqrt{2}$. Both $1 + \sqrt{2}$ and $1 - \sqrt{2}$ are therefore squares, for either choice of the root of 2.

The three differences of the abscissa of R_0 from the roots $0, d, -d$ are

$$1 + \rho, \qquad 2(\rho - 1) = 2/(1 + \rho), \qquad 4.$$

They are nonzero squares modulo p. The elementary full-two-torsion descent criterion for $y^2 = (x - e_1)(x - e_2)(x - e_3)$ therefore puts the point in $2E(\mathbf{F}_p)$. This criterion follows directly from the halving formulas: choose square roots of the three differences with product $-y$ and use their pairwise products to construct the abscissa of a half-point. The same calculation applies to the conjugate.

Finally a finite elliptic group with full rational two-torsion is $\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$ with $a \mid b$ and both a, b even. The exponent of its doubled subgroup is $b/2$, which divides $ab/4 = \#E(\mathbf{F}_p)/4$. □

The inherited simultaneous-prime construction already imposes the residue conditions in this lemma. Thus the existing conjugate order moduli may be used for R_0, not merely for $P = 2R_0$. No additional prime-distribution theorem is required. The exact checker supplies 33 prime examples and a wrong-residue counterexample; these finite calculations supplement, rather than replace, the proof above.

References

[1] J. von Buhren, *Borne de hauteur semi-effective pour le problème de Mordell–Lang dans une variété abélienne*, Journal de Théorie des Nombres de Bordeaux **29** (2017), 307–320, Theorem 1.1, printed p. 308. https://www.numdam.org/article/JTNB_2017__29_1_307_0.pdf.

[2] *Trace heights on quadratic twists with $j = 1728$*, project companion, 13 November 2026, `trace_on_all_points_20261113_v25.tex`.

[3] *A quartic in 204 variables from recovered elliptic differences*, project companion, 14 November 2026, `quartic204_20261114_v27.tex`.